KATO'S EULER SYSTEM

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Contents

1. Goals of this exposé

This is a strict subset of [Kat04]; henceforth, unless otherwise specified, *all references (theorems, equation numbers, subsections, etc.) will be from there.* We also *fix a normalized newform $f \in S_k(\Gamma_1(N))$ for $k \geq 2$, $N \geq 3$.* The goal is to define a certain cohomology class $z(f)$ associated to f and show how it yields special values of L-functions attached to f. A representative statement of this nature, and our first main goal here, is on p.249:

Theorem 1.1. *Take f as above, and say it has Nebentypus ε. Denote $\Omega(f) \in \mathbb{C}^\times$ as the Manin period of f coming from the Eichler–Shimura isomorphism. Let $1 \leq r \leq k/2$, $f^* = \sum_n \overline{a_n} q^n$, and χ be a Dirichlet character. Then, we have*

$$\exp^*(z(f)) = (1 - \overline{a_p} p^{-r} + \overline{\varepsilon}(p) p^{k-1-2r}) \cdot \frac{(2\pi i)^{k-r-1} L(f^*, \chi, r)}{\Omega^\pm(f)} \cdot \iota_{\mathrm{dR},f}.$$

There are two key maps at play. The first is the Eichler–Shimura isomorphism which realizes modular forms in a certain cohomology group of the modular curve. The second is the Bloch–Kato dual exponential map, which in our setting relates the cohomology of the modular curve to modular forms. The rough idea is that, for a fixed f as above, there exists an Euler system of modular forms which produces through the Eichler–Shimura map special L-values in the coefficients, and there is a p-adic Euler system of

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Galois cohomology classes which are sent to the Euler system of modular forms via the dual exponential map. Like how the Galois cohomology classes in the cyclotomic case arise from cyclotomic units via the Kummer map, both Euler systems here arise from Siegel units.

The construction of these two maps are §2 and §3, respectively. The Siegel units and the Euler system of elements in $K_2(Y(N))$ is §4, and the two Euler systems mentioned above are §5 and §6. The two main results from which, for instance, Theorem 1.1 can be deduced are stated in §7. A rough idea of how to obtain Theorem 1.1 is given in §8.

2. Eichler–Shimura isomorphism

We start with an overview of the Eichler–Shimura isomorphism ("per" in Kato). It is our way to pass from modular forms to classes in a certain H^1 of the modular curve. We first give an analytic construction,[1] then give a more cohomological construction.

For some level structure Γ and $k \geq 2$, the map we wish to construct is an isomorphism of Hecke $\mathbb{R}$-modules

$$\mathrm{per} : S_k(\Gamma) \oplus \overline{M_k(\Gamma)} \to H^1(Y(\Gamma), \mathcal{U}_{\mathbb{C}}^{k-2}),$$

where the left hand side are the spaces of cusps and (conjugated) modular forms, $Y(\Gamma) = \Gamma\backslash\mathbb{H}$, and $\mathcal{U}_R^{k-2} = \mathrm{Sym}^{k-2}\,\underline{R}^2$ for a suitably nice ring R, seen as a local system with the usual action of $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ on R^2. (Here we take $R = \mathbb{R}, \mathbb{C}$; in general we also consider $R = \mathbb{Z}, \mathbb{Z}_p$, although we will also switch language a bit.)

Note that we only supply the construction and take the fact that it is an isomorphism of Hecke $\mathbb{R}$-modules for granted. The curious can look here.

The starting point is the holomorphic de Rham complex

$$0 \to \underline{\mathbb{C}} \to \mathcal{O}_{Y(\Gamma)} \to \Omega^1_{Y(\Gamma)} \to 0,$$

which upon tensoring with $\mathcal{U}_{\mathbb{R}}^{k-2}$, noting it is a local system over $\mathbb{R}$ and thus exactness is preserved, gives

$$0 \to \mathcal{U}_{\mathbb{C}}^{k-2} \to \mathcal{U}_{\mathbb{R}}^{k-2} \otimes_{\underline{\mathbb{R}}} \mathcal{O}_{Y(\Gamma)} \to \mathcal{U}_{\mathbb{R}}^{k-2} \otimes_{\underline{\mathbb{R}}} \Omega^1_{Y(\Gamma)} \to 0.$$

Let $\{e_1, e_2\}$ be a basis of $\mathbb{R}^2$. We have a natural map

$$\begin{aligned} M_k(\Gamma) &\to H^0(Y(\Gamma), \mathcal{U}_{\mathbb{R}}^{k-2} \otimes_{\underline{\mathbb{R}}} \Omega^1_{Y(\Gamma)}) \\ f &\mapsto \omega_f(z) \coloneqq (e_1 - ze_2)^{k-2} \otimes f(z)dz, \end{aligned}$$

where $(e_1 - ze_2)^{k-2}$ denotes the product

$$(e_1^{k-2}, e_1^{k-3} \otimes e_2, \cdots, e_2^{k-2}) \begin{pmatrix} & & & 1 \\ & & \ddots & \\ & (-1)^{k-1}\binom{k-2}{1} & & \\ (-1)^{k-2} & & & \end{pmatrix} \begin{pmatrix} z^{k-2} \\ z^{k-3} \\ \vdots \\ 1 \end{pmatrix}.$$

[1] not necessary, and also not given in Kato, but good for concreteness

We can send the target of the above map to $H^1(Y(\Gamma), \mathcal{U}_{\mathbb{C}}^{k-2})$ via the coboundary map δ from de Rham cohomology. Taking their composition gives a natural map

$$\begin{aligned} M_k(\Gamma) &\to H^1(Y(\Gamma), \mathcal{U}_{\mathbb{C}}^{k-2}) \\ f(z) &\mapsto \delta(\omega_f). \end{aligned}$$

Now for the more "algebraic" version. Note that the construction of per is inherently analytic, so although what follows will resemble algebraic geometry, we are really working with analytic spaces throughout.

Let $\lambda = \lambda_\Gamma : \mathcal{E} \to Y(\Gamma)$ be the universal elliptic curve over $Y(\Gamma)$. By standard facts about higher pushforwards of constant sheaves, the sheaf $\mathcal{H}^1_R := R^1\lambda_*(\underline{R})$ is a local system on $Y(\Gamma)$ of rank 2 over R. We define like before $\mathcal{U}_R^{k-2} := \operatorname{Sym}_{\underline{R}}^{k-2} \mathcal{H}^1_R$. We start with the relative Hodge complex

$$0 \to \lambda^*\mathcal{O}_{Y(\Gamma)} \to \mathcal{O}_{\mathcal{E}} \xrightarrow{d} \Omega^1_{\mathcal{E}/Y(\Gamma)} \to 0,$$

which induces the long exact sequence

$$0 \to \lambda_*\lambda^*\mathcal{O}_{Y(\Gamma)} \to \lambda_*\mathcal{O}_{\mathcal{E}} \to \lambda_*\Omega^1_{\mathcal{E}/Y(\Gamma)} \to R^1\lambda_*(\lambda^*\mathcal{O}_{Y(\Gamma)}) \to R^1\lambda_*\mathcal{O}_{\mathcal{E}}.$$

Now we note

- $\lambda_*\lambda^*\mathcal{O}_{Y(\Gamma)} \simeq \lambda_*\mathcal{O}_{\mathcal{E}}$
- $\lambda_*\Omega_{\mathcal{E}/Y(\Gamma)}$ is just the Hodge line bundle ω
- the cup product $R^1\lambda_*\underline{\mathbb{R}} \otimes_{\underline{\mathbb{R}}} \mathcal{O}_{Y(\Gamma)} \simeq R^1\lambda_*\underline{\mathbb{C}} \otimes_{\mathbb{C}} \lambda_*(\lambda^*\mathcal{O}_{Y(\Gamma)}) \xrightarrow{\cup} R^1\lambda_*(\lambda^*\mathcal{O}_{Y(\Gamma)})$ can be checked to be an isomorphism (look on fibers)

so we can rewrite the long exact sequence as

$$0 \to \lambda_*\Omega^1_{\mathcal{E}/Y(\Gamma)} \to R^1\lambda_*\underline{\mathbb{R}} \otimes_{\underline{\mathbb{R}}} \mathcal{O}_{Y(\Gamma)} \to R^1\lambda_*\mathcal{O}_{\mathcal{E}}.$$

On its fibers this just reduces to the Hodge decomposition for elliptic curves, which in particular is a split exact sequence, so the above is in fact exact. By relative Serre duality, we may replace $R^1\lambda_*\mathcal{O}_{\mathcal{E}}$ with $\lambda_*(\Omega^1_{\mathcal{E}/Y(\Gamma)})^\vee = \omega^{-1}$, so we have

$$0 \to \omega \xrightarrow{\iota} R^1\lambda_*\underline{\mathbb{R}} \otimes_{\underline{\mathbb{R}}} \mathcal{O}_{Y_\Gamma} \to \omega^{-1} \to 0.$$

The map ι now induces the $(k-2)$-th tensor power $\iota^{k-2} : \omega^{\otimes(k-2)} \hookrightarrow \mathcal{U}_{\mathbb{R}}^{k-2} \otimes \mathcal{O}_{Y(\Gamma)}$, and thus an inclusion $\omega^{\otimes(k-2)} \otimes \Omega^1_{Y(\Gamma)} \hookrightarrow \mathcal{U}_{\mathbb{R}}^{k-2} \otimes \Omega^1_{Y(\Gamma)}$. Composing this map on global sections with the coboundary map from the de Rham cohomology complex

$$0 \to \underline{\mathbb{C}} \to \mathcal{O}_{Y(\Gamma)} \to \Omega^1_{Y(\Gamma)} \to 0$$

we get

$$\text{per} : H^0(Y(\Gamma), \omega^{\otimes(k-2)} \otimes \Omega^1_{Y(\Gamma)}) \xrightarrow{\iota^{k-2}} H^0(Y(\Gamma), \mathcal{U}_{\mathbb{R}}^{k-2} \otimes \Omega^1_{Y(\Gamma)}) \xrightarrow{\delta} H^1(Y(\Gamma), \mathcal{U}_{\mathbb{C}}^{k-2}).$$

Noting that the first term (with $X(\Gamma)$ instead of $Y(\Gamma)$, but $H^0(X, -) \hookrightarrow H^0(Y, -)$ anyways) is nothing more than the space of cusp forms of weight k and level Γ, we have constructed our Eichler–Shimura map.

Moving forward, we adopt Kato's notation

$$V_{k,R}(Y(\Gamma)) := H^1(Y(\Gamma), \mathcal{U}_R^{k-2}).$$

When R is a $\mathbb{Z}_p$-algebra, then via the usual Betti-to-étale comparison isomorphisms, we have an isomorphism

$$V_{k,R}(Y(\Gamma)) \simeq H^1_{\text{ét}}(Y(\Gamma)_{\overline{\mathbb{Q}}}, \mathcal{U}_R^{k-2}),$$

and so in particular $V_{k,R}(Y(\Gamma))$ can be seen as a representation of $G_{\mathbb{Q}}$ in this case. If $\Gamma = \Gamma(N)$, then the representation is unramified away from pN.

While we're here, we also introduce spaces $V_{k,K_f}(f)$, where $f = \sum a_n q^n \in S_k(\Gamma_1(N))$ is a normalized newform with coefficient field $K_f \subset \mathbb{C}$. Define $V_{k,K_f}(f)$ as the quotient of $V_{k,K_f}(Y_1(N)) = V_{k,\mathbb{Q}}(Y_1(N)) \otimes_{\mathbb{Q}} K_f$ by the K_f-submodule generated by the images of $T_n \otimes 1 - 1 \otimes a_n$ for all n.

We will adorn per with various subscripts to indicate certain level structures or restriction to certain Hecke isotypic parts. We also frequently restrict to just the space of cusp forms. For instance, we denote

$$\mathrm{per}_N : S_k(\Gamma(N)) \to V_{k,\mathbb{C}}(Y(N)),$$

$\mathrm{per}_{1,N}$ for level $\Gamma_1(N)$-structure, and for a choice of $f \in S_k(\Gamma_1(N))$ like above,

$$\mathrm{per}_f : S(f) \to V_{k,\mathbb{C}}(f),$$

where $S(f)$ is defined as the quotient of $M_k(\Gamma_1(N)) \otimes_{\mathbb{Q}} K_f$ by the K_f-submodule generated by the images of the operators $T_n \otimes 1 - 1 \otimes a_n$.

For our two constructions of Euler systems (one consisting of modular forms, the other of Galois cohomology classes), we will have three versions of each: one for level $\Gamma(N)$, then for $\Gamma_1(N)$, then for the f-isotypic projection. This neceesitates the constructions above.

3. Bloch–Kato dual exponential map

This already appeared in the explicit reciprocity map for GL(1) in Yuan Wang's talk, and was essential to produce the Kubota–Leopoldt p-adic zeta function from the Euler system of cyclotomic units. It plays a similarly central role here.

The general construction of $\exp^*$ for a de Rham representation V of G_K, where K is a p-adic field, is the composition

$$\exp^* : H^1(K, V) \to H^1(K, B^0_{\mathrm{dR}} \otimes V) \xleftarrow{\times \log(\chi_{\mathrm{cyc}})} H^0(K, B^0_{\mathrm{dR}} \otimes V) = D^0_{\mathrm{dR}}(V).$$

Take $V = V_{k,\mathbb{Q}_p}(Y(N))(i)$ for $k \geq 2$ and $1 \leq i \leq k-1$. The key fact that makes $\exp^*$ easy to use is that the target is, for any $1 \leq i \leq k-1$,

$$D^0_{\mathrm{dR}}(V) \simeq M_k(\Gamma(N)) \otimes \mathbb{Q}_p,$$

and so $\exp^*$ can be seen as a map

$$\exp^* : H^1(\mathbb{Q}_p, V_{k,\mathbb{Q}_p}(Y(N))(i)) \to M_k(\Gamma(N)) \otimes \mathbb{Q}_p.$$

These are a general consequence of Scholl's construction of motives for f of weight $k \geq 2$ (see §11). We will show the isomorphism for $k = 2$ (and forcibly $i = 1$); the general case works essentially by replacing the modular curve with the appropriate Kuga–Sato variety. We invoke the étale-to-de Rham comparison isomorphism

$$D_{\mathrm{dR}}(H^1_{\text{ét}}(Y(N)_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p)) \simeq H^1_{\mathrm{dR}}(Y(N)) \otimes \mathbb{Q}_p.$$

As this preserves filtrations, we see that

$$\begin{aligned} D^0_{\mathrm{dR}}(H^1_{\text{ét}}(Y(N)_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p)(1)) \simeq D^1_{\mathrm{dR}}(H^1_{\text{ét}}(Y(N)_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p)) &\simeq \mathrm{Fil}^1 H^1_{\mathrm{dR}}(Y(N)) \otimes \mathbb{Q}_p \\ &= H^0(Y(N), \Omega^1_{Y(N)}) \otimes \mathbb{Q}_p = M_2(\Gamma(N)) \otimes \mathbb{Q}_p. \end{aligned}$$

4. Siegel units and K_2

We now construct $z(f)$ for a normalized newform $f \in S_k(\Gamma_1(N))$ which we permanently fix.

Like how the Kubota–Leopoldt p-adic zeta function can be seen as originating from cyclotomic units, Kato's Euler system originates from Siegel units, which are elements of $\mathcal{O}(Y(N))^\times$. In fact, just like how we could form an "Euler system"[2] of cyclotomic units, where the distribution relations are with respect to the corestriction maps, these Siegel units will form elements of $K_2(Y(N))$ which satisfy distribution relations, and these relations will carry over until the end.

The starting point is the following proposition, which we paraphrase for brevity. (See Kato Proposition 1.3) We take it for granted.

Proposition 4.1. *Let E be an elliptic curve over S and $c \in \mathbb{Z}$ with $(c, 6) = 1$. Then, there exists a unique element ${}_c\theta_E \in \mathcal{O}(E \setminus E[c])^\times$ satisfying*

(1) $\operatorname{div} {}_c\theta_E = c^2(0) - E[c]$
(2) invariance under norm maps.

It is also compatible under pullback by different multiplication maps.

Let $\mathcal{E}$ be the universal elliptic curve over $Y(N)$. Recall $Y(N)$ represents the functor which sends a scheme S to isomorphism classes of triples (E, e_1, e_2) where E/S is an elliptic curve and (e_1, e_2) forms a $\mathbb{Z}/N$-basis of $E[N]$. For any $(a/N, b/N) \in (\frac{1}{N}\mathbb{Z}/\mathbb{Z})^2 \setminus \{(0,0)\}$ and c as before, define the map

$$\iota_{a/N,b/N} = ae_1 + be_2 : Y(N) \to \mathcal{E} \setminus \mathcal{E}[c],$$

and then define the **Siegel unit**

$${}_cg_{a/N,b/N} = \iota^*_{a/N,b/N}({}_c\theta_{\mathcal{E}}) \in \mathcal{O}(Y(N))^\times.$$

Now we form elements in $K_2(Y(N))$. Defining K_2 in general is quite difficult, but at least for a field F, it can be defined as

$$K_2(F) \coloneqq F^\times \otimes_{\mathbb{Z}} F^\times / \langle a \otimes (1-a) : a \neq 0, 1 \rangle .$$

So $K_2(Y(N)) \coloneqq K_2(\mathcal{O}(Y(N)))$, which we can view as a subgroup of K_2 of the fraction field of $\mathcal{O}(Y(N))$.

[2]Whenever I put "Euler system" in quotes, I mean a certain family of objects satisfying a certain distribution relation. I will drop the quotes when referring to what is generally known as Kato's Euler system.

For any $c, d \in \mathbb{Z}$ with $(c, 6N) = (d, 6N) = 1$, we define elements of our first "Euler system", valued in $K_2(Y(N))$.[3]

$$ {}_{c,d}z_N = \{{}_c g_{1/N,0}, {}_d g_{0,1/N}\} \in K_2(Y(N)). $$

We state without proof that these are norm-compatible. The proof boils down to taking advantage of the nice properties of ${}_c\theta_E$.

Proposition 4.2 (Proposition 2.3)**.** *For $M \mid N$, the norm homomorphism $K_2(Y(N)) \to K_2(Y(M))$ sends ${}_{c,d}z_N \mapsto {}_{c,d}z_M$.*

From this "Euler system" of elements ${}_{c,d}z_N$ in K_2, we will construct an "Euler system" of modular forms and an Euler system of Galois cohomology classes. Of course, having the same origin, they must be related, and indeed they will essentially occupy opposite sides of $\exp^*$.

5. p-adic Euler system

This is §8.4, nearly verbatim.

Fix $N \geq 3$ and $k \geq 2$, and consider any $r, r' \in \mathbb{Z}$ with $1 \leq r' \leq k-1$. We will define a canonical homomorphism

$$ \mathrm{Ch}_N(k, r, r') : \varprojlim_n K_2(Y(Np^n)) \to H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y(N))(k-r)). $$

Recall from above that $({}_{c,d}z_{Np^n})_{n\geq 1}$ is a norm-compatible system, hence an element of the source of $\mathrm{Ch}_N(k, r, r')$. The elements

$$ {}_{c,d}z_N^{(p)}(k, r, r') \in H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y(N))(k-r)) $$

of our p-adic Euler system will be defined as the image of $({}_{c,d}z_{Np^n})_{n\geq 1}$ under $\mathrm{Ch}_N(k, r, r')$.

The map $\mathrm{Ch}_N(k, r, r')$ on finite levels is the following composition. All steps are defined below.

$$
\begin{aligned}
&(1) && K_2(Y(Np^n)) \to H^2_{\text{ét}}(Y(Np^n)_{\mathbb{Q}}, (\mathbb{Z}/p^n)(2)) \\
&(2) && \to H^2_{\text{ét}}(Y(Np^n)_{\overline{\mathbb{Q}}}, (\mathrm{Sym}^{k-2}(T_p\mathcal{E}_n/p^n))(2-r)) \\
&(3) && \xrightarrow{\sim} H^2_{\text{ét}}(Y(Np^n), (\mathrm{Sym}^{k-2}_{\mathbb{Z}_p}\mathcal{H}^1_p/p^n)(k-r)) \\
&(4) && \to H^2_{\text{ét}}(Y(N), (\mathrm{Sym}^{k-2}_{\mathbb{Z}_p}\mathcal{H}^1_p/p^n)(k-r)) \\
&(5) && \to H^1(\mathbb{Q}, V_{k,\mathbb{Z}/p^n}(Y(N))(k-r)).
\end{aligned}
$$

Each line:

1. Chern character map $K_2(Y(M)) \to H^2(Y(M), (\mathbb{Z}/p^n)(2))$ which sends $\{f, g\} \mapsto \mathrm{Kum}(f) \cup \mathrm{Kum}(g)$.
2. Product with $e_{1,n}^{\otimes(r'-1)} \otimes e_{2,n}^{\otimes(k-r'-1)} \otimes (\zeta_{p^n})^{\otimes -r}$.
   - $\lambda_n : \mathcal{E}_n \to Y(Np^n)$ is the universal elliptic curve, $T_p\mathcal{E}_n$ is the p-adic Tate module seen as a p-adic smooth sheaf on $Y(Np^n)_{\overline{\mathbb{Q}}}$
   - $(e_{1,n}, e_{2,n})$ is a basis of $T_p\mathcal{E}_n/p^n$ over $Y(Np^n)$

[3]Kato works with $Y(M, N)$, but I am lazier than he is (so $M = N$ for me).

- ζ_{p^n} a generator of $(\mathbb{Z}/p^n)(1)$

(3) Invoking isomorphism $T_p\mathcal{E}_n \simeq \mathcal{H}^1_p(1)$ given by Poincaré duality, where recall $\mathcal{H}^1_p = R^1\lambda_{n,*}\underline{\mathbb{Z}_p}$.
(4) Trace map induced by natural cover $Y(Np^n) \to Y(N)$.
(5) From Hochschild–Serre spectral sequence

$$E_2^{i,j} = H^i(\mathbb{Q}, H^j_{\text{ét}}(Y(N)_{\overline{\mathbb{Q}}}, \mathcal{F})) \implies H^{i+j}_{\text{ét}}(Y(N)_{\mathbb{Q}}, \mathcal{F})$$

with the fact that $H^j_{\text{ét}}(Y(N)_{\overline{\mathbb{Q}}}, \mathcal{F}) = 0$ for $j \geq 2$.

Note that the target as stated is in $\varprojlim_n H^1(\mathbb{Q}, V_{k,\mathbb{Z}/p^n}(Y(N))(k-r))$, whereas we really want to land in $H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y(N))(k-r))$. We have a natural injection

$$\begin{aligned} H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y(N))(k-r)) &= \varprojlim_n H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}/p^n}(Y(N))(k-r)) \\ &\hookrightarrow \varprojlim_n H^1(\mathbb{Q}, V_{k,\mathbb{Z}/p^n}(Y(N))(k-r)), \end{aligned}$$

and one can use the fact that $Y(Np^n) \to Y(N)$ factors through $Y(N) \otimes \mathbb{Q}(\zeta_{p^n})$ to deduce that ${}_{c,d}z_N^{(p)}$ in fact lives in the (image of the) desired space. (See Lemma 8.5, §8.6 for details.)

These ${}_{c,d}z_N^{(p)}$ are compatible across different N with respect to the norm map, i.e., they form an "Euler system", although I will not write out the explicit statement. We remark that certain Euler factors pop out in the relations, which manifest in Proposition 5.1.

Now we make everything have level $\Gamma_1(N)$ structure instead. The way to do this is realize $Y_1(N) \otimes \mathbb{Q}(\zeta_m)$ as a canonical projection of $Y(L)$, where both m, N divide L. To descend to $\Gamma_1(N)$ level structure, we require two inputs: (i) $\xi \in \mathrm{SL}_2(\mathbb{Z})$ and (ii) S a finite set of primes containing all primes dividing mN.[4]

Take any such pair (ξ, S). Choose any $L \geq 3$ such that both m, N divide L and $S = \{\ell \mid L\}$. The standard projection $Y(L) \to Y_1(N) \otimes \mathbb{Q}(\zeta_m)$ induces the trace map $V_{k,\mathbb{Q}_p}(Y(L)) \to V_{k,\mathbb{Q}_p}(Y_1(N) \otimes \mathbb{Q}(\zeta_m))$, which also works over $\mathbb{Z}_p$. This gives a homomorphism

$$\begin{aligned} H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y(L))) &\to H^1(\mathbb{Z}[1/p], V_{k,\mathbb{Z}_p}(Y_1(N) \otimes \mathbb{Q}(\zeta_m))) \\ &\cong H^1(\mathbb{Z}[1/p, \zeta_m], V_{k,\mathbb{Z}_p}(Y_1(N))), \end{aligned}$$

and we define ${}_{c,d}z_{1,N,m}^{(p)}(k, r, r', \xi, S)$ to be the image of $\xi^*({}_{c,d}z_L^{(p)}(k, r, r'))$ under this homomorphism. The choice of L in the end does not matter because these ${}_{c,d}z_L^{(p)}$ are norm compatible. These, too, satisfy distribution relations, although again we don't write them out.

Finally, we define ${}_{c,d}z_m^{(p)}(f, r, r', \xi, S) \in H^1(\mathbb{Z}[\zeta_m, 1/p], V_{\mathcal{O}_\lambda}(f))$ for a fixed normalized newform $f \in S_k(\Gamma_1(N))$ of Nebentypus ε to simply be the image of ${}_{c,d}z_{1,N,m}^{(p)}(k, r, r', \xi, S)$

[4]There is another possibility for (ξ, S) in Kato, which is essential for relating the Euler system to special values. However, (a) the notation is already bad as it is, and (b) I only give a sketch of the proof, so this will suffice.

under the natural projection induced by $V_{k,\mathcal{O}_\lambda}(Y_1(N)) \twoheadrightarrow V_{k,\mathcal{O}_\lambda}(f)$. Here, $\mathcal{O} \subset K_f$ and $\mathcal{O}_\lambda$ is the completion at a place λ above p. Once more, these satisfy distribution relations, which we actually write out this time.

Proposition 5.1 (Proposition 8.12)**.** *Keep all notation as above. Let $m' \geq 1$, $m \mid m'$, and let S' be a finite set of primes containing all primes in S and dividing m', but not containing any primes dividing cd. Then, the norm map $H^1(\mathbb{Z}[\zeta_{m'}, 1/p], V_{\mathcal{O}_\lambda}(f)) \to H^1(\mathbb{Z}[\zeta_m, 1/p], V_{\mathcal{O}_\lambda}(f))$ sends the element ${}_{c,d}z^{(p)}_{m'}(f, r, r', \xi, S')$ to*

$$\left(\prod_{\ell \in S'-S} \left(1 - \overline{a_\ell}\sigma_\ell^{-1} \cdot \ell^{-r} + \overline{\varepsilon}(\ell)\sigma_\ell^{-2} \cdot \ell^{k-1-2r}\right)\right) \cdot {}_{c,d}z^{(p)}_m(f, r, r', \xi, S).$$

6. Euler system of modular forms

Now we construct the modular form counterparts to the"zeta elements" defined above. These will be

$$
{}_{c,d}z_N(k, r, r') \in M_k(\Gamma(N)), \qquad {}_{c,d}z_{1,N,m}(k, r, r', \xi, S) \in M_k(\Gamma_1(N)) \otimes \mathbb{Q}(\zeta_m),
$$
$$
{}_{c,d}z_m(f, k, r, r', \xi, S) \in S(f) \otimes \mathbb{Q}(\zeta_m).
$$

The latter two will be obtained from the first, similar to how ${}_{c,d}z^{(p)}_{1,N,m}(k, r, r', \xi, S)$ and ${}_{c,d}z^{(p)}_m(f, r, r', \xi, S)$ were obtained from ${}_{c,d}z^{(p)}_N(k, r, r')$. The first is defined as a product of two Eisenstein series. The motivation for this, in short, is that these "zeta" modular forms produce special L-values essentially from the Rankin–Selberg method, so Eisenstein series are needed to make this possible.

To quote Kato, "Eisenstein series are additive analogues of Siegel units (which are multiplicative elements)." We return to the notation used to define Siegel units. Let $N \geq 3$ and $c \in \mathbb{Z}$ with $(c, 6N) = 1$, and recall the function ${}_c\theta_E \in \mathcal{O}(\mathcal{E} \setminus \mathcal{E}[c])$, where now $\lambda : \mathcal{E} \to Y(N)$ is the universal elliptic curve. We take the "additive" elements

$$\mathrm{dlog}_c\theta_{\mathcal{E}} \in \Gamma(\mathcal{E} \setminus \mathcal{E}[c], \Omega^1_{\mathcal{E}/Y(N)}).$$

Let $\omega_{\mathcal{E}} = \lambda_*\Omega^1_{\mathcal{E}/Y(N)}$ be the usual Hodge line bundle. For $r \in \mathbb{Z}$, we define a derivative $D : \lambda^*\omega_{\mathcal{E}}^{\otimes r} \to \lambda^*\omega_{\mathcal{E}}^{\otimes(r+1)}$ defined locally by $f \otimes \eta^{\otimes r} \mapsto (df/\eta) \otimes \eta^{\otimes(r+1)}$, where η is a local basis of ω, $f \in \mathcal{O}_{\mathcal{E}}$, and $df/\eta \in \mathcal{O}_{\mathcal{E}}$ satisfies $df = (df/\eta) \cdot \eta$.

Recall the map $\iota_{a/N,b/N} = ae_1 + be_2 : Y(N) \to \mathcal{E}/\mathcal{E}[c]$. We define for $k \geq 1$ the weight k Eisenstein series of level $\Gamma(N)$

$${}_cE^{(k)}_{a/N,b/N} \coloneqq \iota^*_{a/N,b/N}(D^{k-1}\mathrm{dlog}_c\theta_{\mathcal{E}}) \in \Gamma(Y(N), \omega_{\mathcal{E}}^{\otimes k}) = M_k(\Gamma(N)).$$

We also define the Eisenstein series for $(a, b) \neq (0, 0)$

$${}_cF^{(h)}_{a/N,b/N} \coloneqq c^2F^{(h)}_{a/N,b/N} - c^{2-h}F^{(h)}_{ca/N,cb/N},$$

where[5]

$$F^{(k)}_{a/N,b/N} \coloneqq N^{-k} \sum_{x,y \in \mathbb{Z}/N\mathbb{Z}} E^{(k)}_{x/N,y/N} \cdot \zeta_N^{bx-ay}.$$

[5]technically, these are defined slightly differently for weight $k = 2$, but I can't be bothered

Now we are ready to define ${}_{c,d}z_N(k,r,r')$. The restrictions for all of these parameters are

(i) $N \geq 3$ and $k \geq 2$,
(ii) $1 \leq r, r' \leq k-1$ and at least one of them is $k-1$,
(iii) $(c,N) = (d,N) = 1$.

Define

$$ {}_{c,d}z_N(k,r,r') = \begin{cases} (-1)^r (r-1)!^{-1} N^{k-2r-2} \cdot {}_cF^{(k-r)}_{1/N,0} \cdot {}_dE^{(r)}_{0,1/N}, & r' = k-1 \\ (-1)^{r'} (k-2)!^{-1} N^{-k} \cdot {}_cE^{(k-r')}_{1/N,0} \cdot {}_dE^{(r')}_{0,1/N}, & r = k-1 \end{cases} \in M_k(\Gamma(N)). $$

One can show that these satisfy distribution relations (compatible with trace maps $M_k(\Gamma(N')) \to M_k(\Gamma(N))$ for $N \mid N'$), but we will not.

We now define ${}_{c,d}z_{1,N,m}(k,r,r',\xi,S)$, which requires the same two inputs (ξ, S) as ${}_{c,d}z^{(p)}_{1,N,m}(k,r,r',\xi,S)$ above. Like before, choose any $L \geq 3$ such that both m, N divide L and S is the set of primes dividing L. We will require c, d to not have any prime divisors in S. The modular form ${}_{c,d}z_{1,N,m}(k,r,r',\xi,S) \in M_k(\Gamma_1(N)) \otimes \mathbb{Q}(\zeta_m)$ is defined as the image of $\xi^*({}_{c,d}z_L(k,r,r'))$ under the trace map induced by the natural projection $X(L) \to X_1(N) \otimes \mathbb{Q}(\zeta_m)$.

Thanks to the trace compatibility of ${}_{c,d}z_N(k,r,r')$, the above construction does not depend on the choice of L, and furthermore these distribution relations carry over to ${}_{c,d}z_{1,N,m}(k,r,r',\xi,S)$.

Finally, we define ${}_{c,d}z_m(f,r,r',\xi,S) \in S(f) \otimes \mathbb{Q}(\zeta_m)$, for $f \in S_k(\Gamma_1(N))$ as before, to be the image of ${}_{c,d}z_{1,N,m}(k,r,r',\xi,S)$ under the obvious projection $M_k(\Gamma_1(N)) \twoheadrightarrow S(f)$.

7. The two main results

In English, these results are

(1) (Theorem 9.7) The p-adic Euler system is related to the Euler system of modular forms by the dual exponential map.
(2) (Theorem 6.6) Upon choosing bases for either side of per_f, one finds the special L-value associated to the dual of f in the coefficients.

The same kind of results also exists for the level $\Gamma(N)$ and $\Gamma_1(N)$ Euler systems that we constructed, e.g., ${}_{c,d}z^{(p)}_{1,N,m}(k,r,r',\xi,S)$ is related to ${}_{c,d}z_{1,N,m}(k,r,r',\xi,S)$ via $\exp^*$ (Theorem 9.6) and the latter yields special L-values through $\mathrm{per}_{1,N}$ (Theorem 5.6). The statements for $\Gamma(N)$ are Theorems 9.5 and 4.6. In fact, the above results are deduced from Theorems 4.6 $\implies$ 5.6 $\implies$ 6.6 and Theorems 9.5 $\implies$ 9.6 $\implies$ 9.7.

In any case, here they are:

Theorem 7.1 (Theorem 9.7). *Maintain the assumptions (i-iii) above for the integers N, k, r, r', c, d, as well as those for (ξ, S). Let $f \in S_k(\Gamma_1(N))$ be a normalized newform, denote F as the coefficient field of f, and choose a place λ of F over p. Then, the map*

$$ \exp^*_f : H^1(\mathbb{Q}(\zeta_m) \otimes \mathbb{Q}_p, V_{F_\lambda}(f)(k-r)) \to S(f) \otimes_F F_\lambda \otimes \mathbb{Q}(\zeta_m) $$

sends the image of ${}_{c,d}z_m^{(p)}(f,r,r',\xi,S)$ *to*

$$ {}_{c,d}z_m(f,r,r',\xi,S) \in S(f)\otimes\mathbb{Q}(\zeta_m) \subset S(f)\otimes_F F_\lambda \otimes \mathbb{Q}(\zeta_m). $$

Theorem 7.2 (Theorem 6.6(1)). *Let* $\chi : (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$, $\xi \in \mathrm{SL}_2(\mathbb{Z})$, $S \supset \{\ell \mid mN\}$ *finite,* $c,d \in \mathbb{Z}$ *with* $c \equiv d \equiv 1 \pmod N$, *and* $r,r' \in \mathbb{Z}$ *with* $1 \le r,r' \le k-1$ *and at least one of them is* $k-1$. *Let* $(u,v) = (r+2-k,r)$ *if* $r' = k-1$ *and* $(k-r',r')$ *if* $r = k-1$. *Let* $\pm = (-1)^{k-r-1}\chi(-1)$. *Then,*

$$ \sum_{b\in(\mathbb{Z}/m)^\times} \chi(b)\mathrm{per}_f\left(\sigma_b({}_{c,d}z_m(f,r,r',\xi,S))\right)^\pm = L_S(f^*,\chi,r)\cdot(2\pi i)^{k-r-1}\cdot\gamma^\pm, $$

where $\gamma = (c^2 - c^u\overline{\chi}(c))(d^2 - d^v\overline{\chi}(d))\delta(f,r',\xi)$.

The $\pm$ is the usual $x^\pm = \frac{1}{2}(1\pm\iota)x$ where ι is complex conjugation.

We define $\delta(f,r',\xi) \in V_F(f)$. Like with the other constructions, $\delta(f,r',\xi)$ will be the image of $\delta_{1,N}(k,r',\xi) \in V_{k,F}(Y_1(N))$, which further is the image of $\xi^*\delta_L(k,r')$ under the trace map $V_{k,F}(Y(L)) \to V_{k,F}(Y_1(N))$ for any $N \mid L$.

So it suffices to define $\delta_N(k,j)$, which we can do in $V_{k,\mathbb{Z}}(Y(N))$. We will actually define it as a homology class; we realize $V_{k,\mathbb{Z}}(Y(N))$ as a homology group via the following isomorphisms.

$$ \begin{aligned} V_{k,\mathbb{Z}}(Y(N)) &\coloneqq H^1(Y(N)(\mathbb{C}), \mathcal{U}_{\mathbb{Z}}^{k-2}) \\ &\cong H_1(X(N)(\mathbb{C}), \{\text{cusps}\}, \mathcal{U}_{\mathbb{Z}}^{k-2}) \\ &\coloneqq H_1(X(N)(\mathbb{C}), \{\text{cusps}\}, \mathrm{Sym}_{\underline{\mathbb{Z}}}^{k-2}\mathcal{H}^1) \\ &\cong H_1(X(N)(\mathbb{C}), \{\text{cusps}\}, \mathrm{Sym}_{\underline{\mathbb{Z}}}^{k-2}(\mathcal{H}om_{\underline{\mathbb{Z}}}(\mathcal{H}^1,\underline{\mathbb{Z}}))). \end{aligned} $$

Denote $\mathcal{F} \coloneqq \mathrm{Sym}_{\underline{\mathbb{Z}}}^{k-2}(\mathcal{H}om_{\underline{\mathbb{Z}}}(\mathcal{H}^1,\underline{\mathbb{Z}}))$. As a class in $H_1(X(N)(\mathbb{C}),\{\text{cusps}\},\mathcal{F})$, we define $\delta_N(k,j)$ to be the image of $e_1^{j-1}e_2^{k-j-1}$ under the natural map

$$ \Gamma((0,1),\varphi^*\mathcal{F}) = H_1([0,1],\{0,1\},\varphi^*\mathcal{F}) \to H_1(X(N)(\mathbb{C}),\{\text{cusps}\},\mathcal{F}). $$

Here, $\varphi : (0,\infty) \to X(N)$ is the natural one, and (e_1,e_2) is the basis of the constant sheaf $\varphi^*(\mathcal{H}om_{\underline{\mathbb{Z}}}(\mathcal{H}^1,\underline{\mathbb{Z}}))$ which on the stalk at $y \in (0,\infty)$ identifies with $(yi,1)$. (We are using the fact that the stalk of the pullback sheaf on $(0,\infty)$ at $y\in(0,\infty)$ is isomorphic to $H_1(\mathbb{C}/(\mathbb{Z}yi+\mathbb{Z}),\mathbb{Z}) = \mathbb{Z}yi+\mathbb{Z}$.)

We will not prove the first result, although it should not be all that surprising because both ${}_{c,d}z_m(f,r,r',\xi,S)$ and its p-adic counterpart are constructed from the "Euler system" of K_2 elements ${}_{c,d}z_N$.

The very simplistic summary of the proof of the second result is that it is a consequence of the Rankin–Selberg method and the fact that the cup product on the right hand side of the Eichler–Shimura isomorphism matches with the Petersson inner product on the left (and both are perfect pairings).

Proof outline of Theorem 6.6(1). The steps are

(1) Work at the full $\Gamma_1(N)$ level structure rather than just the f-isotypic projection. Also assume $\xi = \mathrm{id}$.

(2) Suffices to show that the cup product with $\mathrm{per}_{1,N}(f)$ on both sides agree.
(3) The cup product, interpreted as the Petersson inner product, gives an identity which follows from Rankin–Selberg.
(4) Deduce the case for general (ξ, S) from the trivial one.[6]

Step 1: Note that our result is a direct consequence of the following result when we project to the f-isotypic part.

Theorem 7.3 (Theorem 5.6)**.** *Using the same notation,*

$$\sum_{b\in(\mathbb{Z}/m)^\times} \chi(b)\mathrm{per}_{1,N}\left(\sigma_b({}_{c,d}z_{1,N,m}(k,r,r',\xi,S))\right)^{\pm} = Z_{1,N,S}(k,\chi,r)\cdot(2\pi i)^{k-r-1}\cdot\gamma^{\pm},$$

where

$$Z_{1,N,S}(k,\chi,s) = \sum_{(n,S)=1}\chi(n)\overline{T_n}n^{-s}, \quad \gamma = (c^2 - c^u\overline{\chi}(c))(d^2 - d^v\overline{\chi}(d))\delta_{1,N}(k,r',\xi).$$

We now proceed with $\xi = \mathrm{id}$ and S being the set of primes dividing m. Recall in our assumptions that one of r, r' must be $k-1$; we assume here $r' = k-1$, although the other case is essentially symmetric. In this case, we have an explicit description of the modular forms appearing on the left hand side.

$$\sigma_a({}_{c,d}z_{1,N,m}(k,r,r',\mathrm{id},S)) = (-1)^r(r-1)!^{-1}N^{-r}m^{k-r-2}\sum_{b\in\mathbb{Z}/m} {}_cF^{(k-r)}_{a/m,b/m}\cdot {}_dE^{(r)}_{0,1/N}.$$

We now define ${}_dz_\chi(r,r') \in M_k(\Gamma_1(N))$ as

$$ {}_dz_\chi(r,r') \coloneqq (-1)^r(r-1)!^{-1}N^{-r}\cdot F^{(k-r)}_\chi \cdot {}_dE^{(r)}_{0,1/N},$$

where

$$F^{(h)}_\chi \coloneqq m^{h-2}\sum_{\substack{a\in(\mathbb{Z}/m)^\times\\ b\in\mathbb{Z}/m}}\chi(a)F^{(h)}_{a/m,b/m}.$$

By definition, the left hand side of the theorem is just $(c^2 - c^u\overline{\chi}(c))\cdot\mathrm{per}_{1,N}(z_\chi(r,r'))$.

Finally, we rewrite the theorem to avoid the $\pm$. Assuming $\chi(-1) = (-1)^{k-r}$, our theorem now looks like

$$\begin{aligned}&\frac{1}{2}\left(\mathrm{per}_{1,N}({}_dz_\chi(r,r')) + (-1)^{k-r-1}\iota(\mathrm{per}_{1,N}({}_dz_{\overline{\chi}}(r,r')))\right)\\ &\quad= Z_{1,N}(k,\chi,r)\cdot(2\pi i)^{k-r-1}\left(d^2 - d^j\overline{\chi}(d)\left(\begin{smallmatrix}1/d & 0\\ 0 & d\end{smallmatrix}\right)^*\right)\cdot\delta_{1,N}(k,r').\end{aligned}$$

Step 2: We apply the cup product $\left\langle -, \mathrm{per}_{1,N}(f)\right\rangle$ on both sides. On the left, we note that the cup product corresponds via the Eichler–Shimura isomorphism to the

[6]Kato's proof actually does not treat the case when $\xi \in \mathrm{SL}_2(\mathbb{Z})$, although it is certainly part of the statement of Theorem 6.6. The proof for the other case of (ξ, S) is done by first treating the trivial case, then deducing everything else from the trivial case. I am not sure exactly how to proceed in our situation...

Petersson inner product. On the right, by construction we get

$$\begin{aligned}\left\langle \delta_{1,N}(k,r'), \mathrm{per}_{1,N}(f)\right\rangle &= (2\pi i)^{k-1}\int_0^\infty f(iy)\cdot(iy)^{j-1}d(iy)\\ &= (2\pi i)^{k-j-1}(-1)^j(j-1)!\cdot L(f,j).\end{aligned}$$

Applying the operator-valued zeta function $Z_{1,N}(k,\chi,r)$ gives us a factor of $L(f^*,\chi,r)$.

Step 3: Writing everything out, the equality we are trying to prove should look like

$$\begin{aligned}&\frac{1}{2}(-8\pi^2 i)^{k-1}\int_{\Gamma_1(N)\backslash\mathbb{H}}\overline{f(\tau)}z_\chi(r,r')y^k\frac{dxdy}{y^2}\\ &\qquad\stackrel{?}{=}(2\pi i)^{2k-r-r'-2}(-1)^{r'}(r'-1)!L(f^*,\chi,r)L(f^*,r').\end{aligned}$$

But, with some additional information on the coefficients of $F_\chi^{k-r'}$ (see Proposition 3.10), this falls directly from the Rankin–Selberg method. □

8. Extracting special values

We apply the two main results to get from Kato's zeta elements to special L-values. The statement we want is Theorem 12.5 (where the L-value is $L_{\{p\}}(f^*,\chi,r)$), but we will "prove" a weaker version that produces $L_S(f^*,\chi,r)$. Shrinking S to just $\{p\}$ is not difficult, but it requires some external results and the proof is not particularly enlightening. We conclude by showing how Theorem 12.5 can get us a result like the Theorem 1.1 in the introduction of this exposé.

Fix once again a normalized newform $f=\sum a_nq^n\in S_k(\Gamma_1(N))$ with $k\geq 2$ and coefficient field F. Let $\Lambda=\mathbb{Z}_p[\![\mathrm{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q})]\!]$ be the usual Iwasawa algebra and $Q(\Lambda)$ its total quotient ring. Define

$$H^1_{\mathrm{Iw}}(V_{F_\lambda}(f)) := \varprojlim_n H^1(\mathbb{Q}(\zeta_{p^n}),V_{F_\lambda}(f)).$$

We now define, for $\gamma\in V_{F_\lambda}(f)$, a certain p-adic zeta element $\mathbf{z}_\gamma^{(p)}$. Our construction will define it as an element of $H^1_{\mathrm{Iw}}(V_{F_\lambda}(f))\otimes_\Lambda Q(\Lambda)$, although one can in fact prove that it is integral over Λ. (This is one part of Theorem 12.5 which we do not prove.)

We choose elements $\xi_1,\xi_2\in\mathrm{SL}_2(\mathbb{Z})$ and $1\leq j_1,j_2\leq k-1$ such that $\delta(f,j_1,\xi_1)^+$ and $\delta(f,j_2,\xi_2)^-$ are both nonzero. Such choices (j_1,j_2,ξ_1,ξ_2) must exist because $V_{k,\mathbb{Z}}(Y(N))$ is generated over $\mathbb{Z}$ by all $\xi^*\delta_N(k,j)$ by Ash–Stevens.

Note by the Multiplicity One theorem and the fact that per_f is an isomorphism, we have $\dim_{F_\lambda}V_{k,F_\lambda}(f)^\pm=1$. Thus, for any $\gamma\in V_{k,F_\lambda}(f)$, there exists $\alpha_1,\alpha_2\in F_\lambda$ such that

$$\gamma=\alpha_1\cdot\delta(f,j_1,\xi_1)^+ + \alpha_2\cdot\delta(f,j_2,\xi_2)^-.$$

We now define, for $c, d \in \mathbb{Z}$ satisfying $(cd, 6p) = 1$, $c \equiv d \equiv 1 \pmod{N}$, and $c, d \neq \pm 1$, the element $\mathbf{z}_\gamma^{(p)} \in H^1_{\mathrm{Iw}}(V_{F_\lambda}(f)) \otimes_\Lambda Q(\Lambda)$. Let S be the set of primes dividing pN.

$$\mathbf{z}_\gamma^{(p)} := \left(\mu(c,d,j_1)^{-1} \cdot \alpha_1 \left({}_{c,d}z_{p^n}^{(p)}(f,k,j_1,\xi_1,S)\right)_{n\geq 1}\right)^+ + \left(\mu(c,d,j_2)^{-1} \cdot \alpha_2 \left({}_{c,d}z_{p^n}^{(p)}(f,k,j_2,\xi_2,S)\right)_{n\geq 1}\right)^-,$$

where

$$\mu(c,d,j) := (c^2 - c^{k+1-j}\sigma_c)(d^2 - d^{j+1}\sigma_d) \prod_{p\neq \ell | N} (1 - \overline{a_\ell}\ell^{-k}\sigma_\ell^{-1}) \in \Lambda.$$

Inspired by the two main results, we want to pass this through $\exp^*$, then per_f. This is done first by the following composition:

$$\begin{aligned}\exp^*_{n,p}(k-r) : H^1_{\mathrm{Iw}}(V_{F_\lambda}(f)) &\xrightarrow{\mathrm{proj}_n} H^1(\mathbb{Q}(\zeta_{p^n}), V_{F_\lambda}(f)) \\ &\xrightarrow{\mathrm{loc}_p} H^1(\mathbb{Q}_p(\zeta_{p^n}), V_{F_\lambda}(f)) \\ &\xrightarrow{\mathrm{twist}} H^1(\mathbb{Q}_p(\zeta_{p^n}), V_{F_\lambda}(f)(k-r)) \\ &\xrightarrow{\exp^*} S(f) \otimes_F F_\lambda \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{p^n}).\end{aligned}$$

At this point, we prove that the image of $\mathbf{z}_\gamma^{(p)}$ under this map is actually in $S(f) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{p^n})$, from which we can then further send it along the map

$$\begin{aligned}\mathrm{per}_{f,\chi} : S(f) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{p^n}) &\to V_{\mathbb{C}}(f)^\pm \\ x \otimes y &\mapsto \sum_{\sigma \in G_n} \chi(\sigma)\sigma(y)\mathrm{per}_f(x)^\pm,\end{aligned}$$

where $G_n = \mathrm{Gal}(\mathbb{Q}(\zeta_{p^n})/\mathbb{Q})$ and $\chi : G_n \to \mathbb{C}^\times$ is a Dirichlet character.

Proposition 8.1. *Choose some $\gamma \in V_{F_\lambda}(f)$, and let $\mathbf{z}_\gamma^{(p)}$ be defined as above. Denote $\mu = \mu(c,d,j_1)\mu(c,d,j_2)$. Then,*

1. *The image of $\mathbf{z}_\gamma^{(p)}$ under the map $H^1_{\mathrm{Iw}}(V_{F_\lambda}(f)) \otimes_\Lambda \Lambda[\mu^{-1}] \to S(f) \otimes_F \overline{F}_\lambda$ induced by $\exp^*_{n,p}(k-r)$ is contained in $S(f) \otimes_F \overline{\mathbb{Q}} \subset S(f) \otimes_F \overline{F}_\lambda \otimes_F \overline{\mathbb{Q}}$.*
2. *Its image along $\mathrm{per}_{f,\chi}$ is exactly $L_S(f^*, \chi, r) \cdot \gamma^\pm$, where again S is the set of primes dividing pN and $\pm = (-1)^{k-r-1}\chi(-1)$.*

Proof. (1) follows from Theorem 9.7. (2) follows from Theorem 6.6. □

As mentioned before, one can do some work to ensure $S = \{p\}$. The other type of pair (ξ, S) which we have not discussed here is crucial for this step, although in the end, the proof boils down to "work carefully" and use the two main results. We state the stronger result below.

Theorem 8.2 (Theorem 12.5(1)). *The map $V_{F_\lambda}(f) \to H^1_{\mathrm{Iw}}(V_{F_\lambda}(f))$ sending $\gamma \mapsto \mathbf{z}_\gamma^\pm$ is F_λ-linear and is the unique map such that*

1. $\exp^*_{n,p}(k-r)(\mathbf{z}_\gamma^{(p)}) \in S(f) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{p^n}) \subset S(f) \otimes_F F_\lambda \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{p^n})$

(2) $\mathrm{per}_{f,\chi} \circ \exp^*_{n,p}(k-r)(\mathbf{z}_\gamma^{(p)}) = (2\pi i)^{k-r-1} \cdot L_{\{p\}}(f^*, \chi, r) \cdot \gamma^\pm$, *where* χ *and* $\pm$ *are as above.*

Using this, proving something like Theorem 1.1 boils down to a wise choice of $\gamma \in V_{F_\lambda}(f)$. Consider again the (twisted) Eichler–Shimura map $\mathrm{per}_f : S(f) \to V_{\mathbb{C}}(f)(k-r)$. Choose an F-basis $\gamma^\pm$ for each of $V_F(f)(k-r)$ as well as $f \in S(f)$. Then, the Manin periods $\Omega^\pm \in \mathbb{C}^\times$ satisfy

$$\mathrm{per}_f(f)^\pm = \Omega^\pm \cdot \gamma^\pm.$$

From the above theorem, we see that for $\gamma = \gamma^+ + \gamma^-$, we have

$$\mathrm{per}_{f,\chi} \circ \exp^*_{n,p}(k-r)(\mathbf{z}_\gamma^{(p)}) = (2\pi i)^{k-r-1} \cdot L_{\{p\}}(f^*, \chi, r) \cdot \gamma^\pm.$$

Defining $z(f)$ as the image of $\mathbf{z}_\gamma^{(p)}$ under $\exp^*_{n,p}(k-r)$ just before the dual exponential (hence an element in $H^1(\mathbb{Q}_p(\zeta_{p^n}), V_{F_\lambda}(f)(k-r))$), we get

$$\begin{aligned}\exp^*(z(f)) &= \frac{(2\pi i)^{k-r-1} \cdot L_{\{p\}}(f^*, \chi, r)}{\Omega^\pm} \cdot f \\ &= (1 - \overline{a_p} p^{-r} + \overline{\varepsilon}(p) p^{k-1-2r}) \frac{(2\pi i)^{k-r-1} \cdot L(f^*, \chi, r)}{\Omega^\pm} \cdot f,\end{aligned}$$

thus recovering Theorem 1.1.

Remark 8.3. Kato does it differently, instead defining γ integrally through theorems of Lafaille. This allows us to define a singular Ω. See §14.17-21.

References

[Kat04] Kazuya Kato. "p-adic L-functions and zeta elements of modular forms". In: *Astérisque* 295 (2004). Cohomologies p-adiques et applications arithmétiques. III, pp. 117–290.